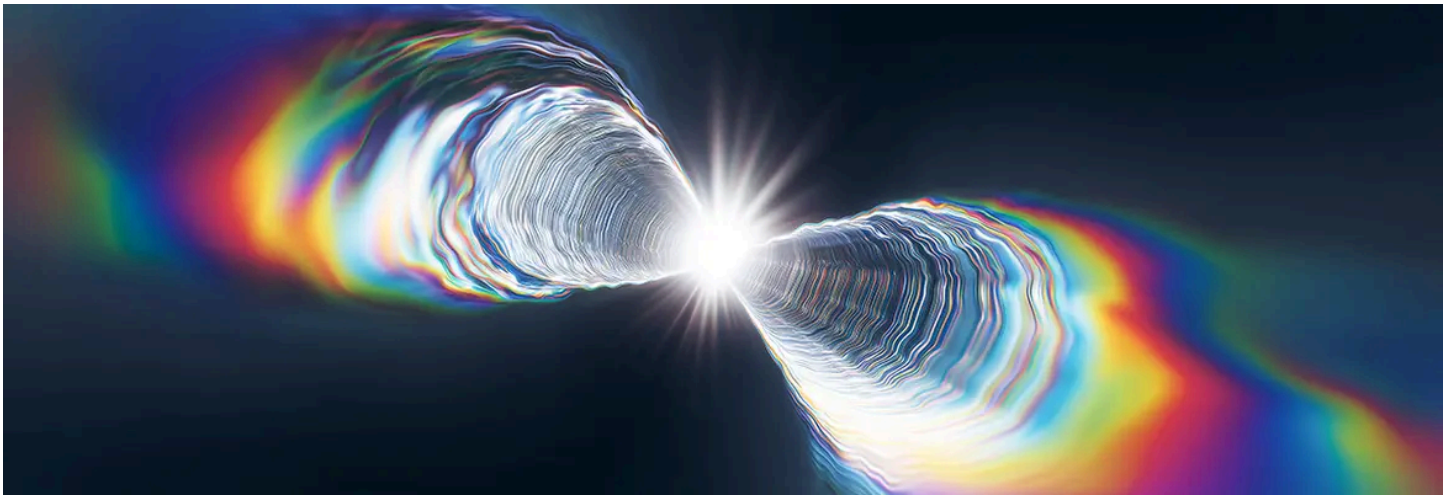


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To unify relativity and quantum mechanics we must abandon materialism

There is something more fundamental than both mind and matter



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 **cite**

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Physicists have so far failed to unify general relativity and quantum mechanics. As attempts to unite them into a quantum theory of gravity mount up, philosopher of physics Dean Rickles argues that the assumption of materialism is the problem. We need to look beyond the physical—beyond space, time and matter—to something primordial out of which minds can construct physical reality, and which explains both general relativity and quantum mechanics. Pioneers like John Wheeler and David Bohm have already begun to chart what such a realm of “pre-physics” might look like—it’s high time physics took their ideas more seriously.

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A pair of recent physics Nobel prizes (2020 and 2022) were awarded for basic research in general relativity (Einstein’s theory of gravitation that explains gravity as

the curvature of spacetime by matter and energy) and quantum mechanics (our best bet for a theory *of matter and energy*). The experimental successes of these theories keep piling up. There is clearly much truth in them. They both aim to describe the same world: *this* world. They should surely overlap, since the matter and energy described by quantum mechanics should curve spacetime as well as good old-fashioned non-quantum mechanical matter and energy. Why then can we not construct a theory in which they both appear? Why is it [so difficult to build what would be a Quantum Theory of Gravity](#)?

It is the biggest scandal in physics, but it is rarely presented as anything other than an amusing paradox or an interesting challenge.

It seems, in crucial ways, their *pictures* of the world are completely at odds. Quantum theory speaks of particles and fields in a *fixed* spacetime that stands quite apart from them. General relativity speaks of a spacetime that gregariously joins in the drama with particles and fields. Demanding the particles and fields obey quantum mechanics spoils their passionate tango with gravity and spacetime. Making the spacetime move and curve by the matter and energy described by quantum theories, spoils the tight rules of quantum theory that make it go. We are as a result in [a more-than-century-old stalemate](#). And consider what this means for our scientific worldview: if these two theoretical frameworks cannot peacefully co-exist, then there is no solid ground for physics to stand on at a fundamental level. It is the biggest scandal in physics, but it is rarely presented as anything other than an amusing paradox or an interesting challenge. It's a mess!

The shackles of orthodox, materialist approaches to doing physics must at the very least be loosened, if not broken off

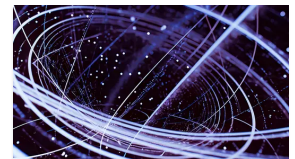
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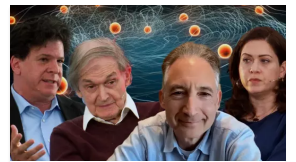
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All rational analysis departs from axioms. The crux of physical analysis is inferring the axioms from observation. When contradictions arise, we are required to isolate our axioms and evaluate how they might be modified to expand our understanding.

The contradictions that divide quantum mechanics (and particularly field theory) from general relativity resolve to a small number of axioms. Committed to the principle that there was no preferred rest frame, Einstein asserted that if we remove all matter, space would be empty. A corollary is smooth fields that can be modeled with differential equations. However, the axiom itself has been disproven by dark energy.

Field theorists hold the axiom that particles are unitary (they have no constituent parts). This is challenged by the regularity of the particle generations, suggesting that they are composed of a structure with excitation states. If we further allow that charge is detachable from that structure, we come immediately to a dualist model of reality.

That the neutrinos have no mass demands that the fundamental (charge-free) structure of the particles must be the building block of the dark energy field, which is recognized as the "aluminiferous ether" that Einstein consigned to intellectual oblivion.

These ideas have been elaborated with a conceptual model whose degrees of freedom can be mapped to all of known physics. The answers are all available. The discipline is simply incurious. I would post a link but the engine disallows that. The motivations can be found be searching for a blog post titled "Einstein Led Us into Mass Hallucinations."

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